SAP S/4HANA MM Configuration Portfolio

Enterprise Structure · Master Data · Procurement · Valuation · Inventory

Düsseldorf Manufacturing AG (DMAG) — Fictional Sandbox Project

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Disclaimer: This is a fictional, sandbox-based SAP S/4HANA MM practice implementation. DMAG is a fictional company used solely to illustrate configuration concepts.

This configuration portfolio is a structured, hands-on SAP S/4HANA MM implementation created to demonstrate my functional consulting capability across enterprise structure, master data, procurement, valuation, inventory, and FI-MM integration. It complements the real-world consulting and operations experience outlined in my CV and showcases my ability to design, configure, test, and document end-to-end MM processes in a manufacturing scenario aligned to DACH-region business practices.

1. Company Background

Company: Düsseldorf Manufacturing AG (DMAG)

Industry: Precision Engineering · Automotive Components · Industrial Equipment Manufacturing

Headquarters: Düsseldorf, North Rhine-Westphalia, Germany

Company Code Currency: EUR | Group Currency: USD

Plants

Plant Code	Location	Purpose
DM01	Düsseldorf	Headquarters and Central Procurement
DM02	Dortmund	Production Plant
DM03	Stuttgart	Assembly and Quality Assurance

Storage Locations

- SL01 — Raw Materials
- SL02 — Semi-Finished Materials
- SL03 — Finished Goods

Purchasing Organisation

- DMPO — Central Purchasing Organisation (assigned to all three plants)
- DMG — Procurement Specialist Purchasing Group

2. Enterprise Structure Configuration

The enterprise structure defines how DMAG is represented in SAP from company level down to storage location. All procurement and inventory transactions reference this structure.

Object	Code / Value	T-Code	Detail
Company	DMAG	OX15	Düsseldorf Manufacturing AG
Company Code	DM01	OX02	Currency: EUR (company) / USD (group) — OBY6
Plant 1	DM01	OX10	Düsseldorf — HQ and Procurement
Plant 2	DM02	OX10	Dortmund — Production
Plant 3	DM03	OX10	Stuttgart — Assembly and QA
Storage Loc	SL01–SL03	OX09	Created under each plant
Purchase Org	DMPO	OX08	Central — assigned to company code OX01 and all plants OX17
Purchase Group	DMG	OME4	Procurement Specialist group

SAP System Evidence *T-Code: OX10 / OX18 / OX17*	
Navigation: *SPRO → Enterprise Structure → Definition → Logistics General → Define Plant*	
Company Code:	`DM01	Currency: EUR	Group Currency: USD`
Plants:	`DM01 Düsseldorf	DM02 Dortmund	DM03 Stuttgart`
Assignment:	`All three plants assigned to company code DM01 via OX18`
Purchase Org:	`DMPO assigned to company code via OX01 and to all plants via OX17`
Storage Locs:	`SL01 Raw	SL02 Semi-Finished	SL03 Finished — created under each plant`
What to verify: *Three plants visible in enterprise structure. DMPO linked to all plants. Storage locations are present under each plant.*	

3. Material Master Configuration

Material master is the central data repository for all materials. Configuration covers material types, number ranges, valuation classes and field selection.

3.1 Material Types and Valuation Classes

Material Type	Code	Valuation Class	Number Range	Sample Material
Raw Material	ROH	3000	200001–200999	200001 — Steel Sheet
Semi-Finished	HALB	7920	300001–300999	300001 — Machined Housing
Finished Goods	FERT	7900	400001–400999	400001 — Pump Assembly

3.2 Key Configuration Steps

Configuration	T-Code	Detail
Number Ranges	MMNR	Internal number ranges defined per material type
Field Selection	OMS9	Required, optional and display fields configured per screen per material type
Material Groups	OMSF	Groups for procurement categorization and spend reporting
Valuation Classes	OMS2	Valuation class assigned per material type — drives OBYC account determination
Delivery Tolerance	OME1	Purchasing Value Key: 10% over-delivery, 10% under-delivery

3.3 Material Master Views Activated

All materials extended to plants DM01, DM02 and DM03 with views: Basic Data 1 and 2, Classification, Purchasing, MRP 1–4, General Plant/Storage, Sales and Accounting.

SAP System Evidence *T-Code: MM01 / MM03*	
Navigation: *Logistics → Materials Management → Material Master → Material → Create (General)*	
Material:	`200001 — Steel Sheet	Type: ROH	Base Unit: KG`
Valuation Class:	`3000 — Raw Material`
Price Control:	`V (Moving Average Price)`
Purchasing View:	`Purch Group: DMG	Delivery tolerance via purchasing value key`
MRP View:	`MRP Type: PD	Lot Size: EX	Safety Stock configured`
Plants Extended:	`DM01 Düsseldorf	DM02 Dortmund	DM03 Stuttgart`
What to verify: *Material 200001 visible in MM03 with all views active. Extended to all three plants.*	

4. Business Partner — Vendor Master

In SAP S/4HANA, vendor data is managed through the Business Partner (BP) framework, replacing the legacy vendor master approach.

BP Role	Code	Purpose
General Data	0000	Name, address, communication — applies to all BP types
Supplier (Company Code)	FLVN00	Reconciliation account, payment terms, payment method
Supplier (Purchasing)	FLVN01	Purchase org data, order currency EUR, incoterms, GR-based IV flag

Account Group	Number Range	Vendor Type
L001 — Domestic Vendors	4000–4499	German and domestic EU suppliers
L002 — EU Vendors	4500–4999	International EU-based suppliers

SAP System Evidence *T-Code: BP*	
Navigation: *Accounting → Financial Accounting → Accounts Payable → Master Records → Maintain Centrally*	
BP Number:	`VEND-DM01 | Name: Steel Supplies GmbH`
Account Group:	`L001 — Domestic Vendor`
BP Roles:	`0000 General | FLVN00 Company Code | FLVN01 Purchasing`
Company Code:	`DM01 | Recon. Account: 160000 | Payment Terms: 30 days net`
Purchase Org:	`DMPO | Order Currency: EUR | GR-Based IV: Active`
What to verify: *All three BP roles assigned to VEND-DM01. Company code and purchasing org data visible.*	

5. Pricing Procedure — Condition Technique

Custom pricing procedure ZDM01 configured for DMAG using the four-step condition technique framework.

Step	Object	Code	T-Code	Configuration
1	**Condition Table**	700	M/03	Fields: Purch Org, Vendor, Material — basis for price lookup
2	**Access Sequence**	ZDM1	M/07	Single access step using condition table 700
3	**Condition Types**	PB00 / RA01 / FRB1	M/06	PB00 Gross Price | RA01 Discount % | FRB1 Freight
4	**Calculation Schema**	ZDM01	M/08	Gross Price → Discount → Freight → Net Value (NETWR)

Calculation Example — Steel Sheet 1,000 KG

Condition	Description	Rate	Amount (EUR)
PB00	Gross Price	2.50 EUR/KG	2,500.00
RA01	Volume Discount (5% for qty > 500 KG)	-5%	-125.00
FRB1	Freight Charges (fixed)	100 EUR flat	100.00
NETWR	Net Order Value	—	2,475.00

6. PO Release Strategy

Classification-based release strategy configured using four characteristics to determine approval levels for purchase orders.

6.1 Characteristics

Characteristic	SAP Field	Values
Plant	WERKS	DM01, DM02, DM03
Purchasing Organization	EKORG	DMPO
Document Type	BSART	NB (Standard PO), UB (STO), MK (Contract)
Net Order Value	GNETW	Value thresholds in EUR — four approval bands

6.2 Approval Levels

Net Order Value (EUR)	Required Approver	Release Code
0 to 50,000	Buyer	01
50,001 to 100,000	Buyer + Senior Buyer	01 + 02
100,001 to 200,000	Buyer + Senior Buyer + Procurement Manager	01 + 02 + 03
Above 200,000	All above + Finance Controller	01 + 02 + 03 + 04

Class: DM_CLASS_PORS | Class Type: 032 | Configured via CL02 (class) and CT04 (characteristics)

SAP System Evidence *T-Code: CL02 / CT04 / OMGSM*	
Navigation: *SPRO → Materials Management → Purchasing → Purchase Order → Release Procedure for Purchase Orders*	
Release Class:	`DM_CLASS_PORS	Class Type: 032`
Release Group:	`01	Codes: 01 Buyer	02 Sr Buyer	03 Mgr.	04 Finance`
Strategy Example:	`PO above 200,000 EUR — all four codes required before issue`
Workflow:	`ME29N for release	SAP Fiori Manage Workflows app also configured`
What to verify: *Release strategy active on document type NB. Test PO above 200,000 EUR shows blocked status pending all four approvals.*	

7. Material Ledger Configuration

Material Ledger is mandatory in SAP S/4HANA. Activated across all three DMAG plants with two parallel currencies.

Step	Activity	T-Code	DMAG Setting
1	**Define ML Type**	OMX2	ML Type ZDM_ML — price determination: actual costing
2	**Assign ML Type to Valuation Area**	OMX3	ZDM_ML assigned to DM01, DM02, DM03
3	**Activate Material Ledger**	CKMSTART	ML activated — status PRODUCTIVE for all plants
4	**Configure Currencies**	OBY6	Type 10: EUR (Company Code) | Type 30: USD (Group)
5	**Actual Costing Run**	CKMLCP	Period-end costing run — actual cost calculated and posted
6	**Material Price Analysis**	CKM3N	Review actual vs standard cost per material per period

8. OBYC — Automatic Account Determination

OBYC maps goods movements to FI GL accounts automatically. Every stock movement generates an accounting document based on the transaction key and valuation class.

Key	Description	Triggered By	GL Account — DMAG
BSX	Inventory Posting	GR (101), GI, STO, transfers	Val 3000→130000 Raw | Val 7920→131000 HALB | Val 7900→132000 FG
WRX	GR/IR Clearing	GR against PO (credit) | MIRO posting (debit)	191100 — GR/IR Clearing Account
PRD	Price Differences	Invoice price ≠ PO price (standard price materials)	281000 — Purchase Price Variance
GBB	Offsetting GI Entries	GI to production (261), cost center (201), scrap	GBB-VBR→400000 Consumption | GBB-VBO→401000
FRE	Freight Clearing	Freight condition posting at GR	410000 — Freight Clearing
KON	Consignment Payable	Withdrawal from consignment stock (201, 261)	160000 — Consignment Payable
INV	Inventory Differences	Physical inventory difference posting (MI07)	170000 — Inventory Difference

9. MRP Configuration

MRP generates procurement proposals automatically based on demand signals. MRP Live (MD01N) is the S/4HANA embedded engine.

MRP Parameter	DMAG Setting	T-Code
MRP Type	PD — Standard MRP (demand-based) | VB — Reorder Point	MM01 MRP 1 view
Lot Sizing	EX — Lot-for-Lot (exact shortfall quantity)	MM01 MRP 1 view
Safety Stock	Maintained on MRP 2 view — triggers PR before depletion	MM01 MRP 2 view
Plant Independent Req.	Demand entered across planning periods for HALB and FERT	MD61
MRP Live Execution	MD01N — S/4HANA embedded MRP, replaces classic MD01/MD02	MD01N
Results Review	Stock/requirements list — planned orders and PRs visible	MD04
Auto PO Conversion	Info Record and Source List prerequisites — ME59N batch conversion	ME01, ME11, ME59N

10. Special Procurement Scenarios

10.1 Stock Transport Order (STO) — DM02 to DM03

Step	Action	T-Code / Movement
1	**Create STO**	ME21N — document type UB, supplying plant DM02, receiving plant DM03

2	**Goods Issue — Supplying Plant**	MIGO — movement type 351, stock moves to in-transit
3	**Stock in Transit**	MMBE — in-transit stock visible at both plants during transfer
4	**Goods Receipt — Receiving Plant**	MIGO — movement type 101, stock updated at DM03 SL02

10.2 Consignment Process

- Consignment PO created with item category K (ME21N)
- GR posts to consignment stock type — no FI accounting document at receipt
- Consignment stock visible in MBLB (vendor-owned stock at plant)
- Withdrawal from consignment (movement type 201) — KON account key creates vendor payable
- Periodic settlement via MRKO — consignment payable cleared to vendor account

10.3 Subcontracting — Pump Assembly

- BOM created (CS01) for Pump Assembly (400001) — child components defined
- Purchase Info Record created for subcontracting processing charges
- Subcontracting PO with item category L (ME21N)
- Components transferred to subcontractor: MIGO movement type 541 — special stock O
- Finished Pump Assembly received: MIGO movement type 101 — components auto-consumed via 543

11. Batch Management and Split Valuation

11.1 Batch Management — Steel Sheet

Step	Activity	T-Code
1	**Activate Batch Management at Plant Level**	OMCE
2	**Create Batch Class (Class Type 023)**	CL01 — Class: DM_BATCH_CLASS
3	**Create Batch Characteristics**	CT04 — Characteristic: DM_GRADE | Values: Grade-A, Grade-B, Grade-C
4	**Assign Class to Material**	MM01 Classification view — class assigned to material 200001
5	**Create Batch Record at GR**	MSC1N — batch created with grade characteristic

11.2 Split Valuation — Steel Sheet by Quality Grade

Valuation Type	Category	Standard Price (EUR/KG)	GL Account (BSX)
GRADE-A	Quality Grade A — Premium	3.20	130001 Premium Steel
GRADE-B	Quality Grade B — Standard	2.50	130000 Standard Steel
GRADE-C	Quality Grade C — Economy	1.80	130002 Economy Steel

Split valuation activated via OMWC. Separate accounting views maintained per valuation type.

12. Physical Inventory

Full physical inventory cycle executed at plant DM01, storage location SL01.

Step	Activity	T-Code	Outcome
1	**Define PI Number Range**	OMBT	Number range assigned to PI document type
2	**Create PI Document**	MI01	PI document created for DM01/SL01 — all ROH materials included

3	**Enter Counted Quantities**	MI04	Physical count results entered per material line
4	**Review Differences**	MI20	Difference list: book quantity vs counted quantity
5	**Post Differences**	MI07	Stock adjusted, FI document via OBYC key INV

13. Testing and Validation

Test Area	Test Scenario	Expected Result
Pricing	Create PO 1,000 KG Steel Sheet from VEND-DM01	PB00: 2,500 | RA01: -125 | FRB1: 100 | Net: 2,475 EUR
Release Strategy	Create PO with value 250,000 EUR	PO blocked — all four release codes required
P2P Cycle	Execute full PR to MIRO and verify FI postings	BSX and WRX at GR | WRX cleared and vendor credited at MIRO
Split Valuation	GR for Grade-A and Grade-B batches separately	Separate stock quantities and separate GL accounts posted
Physical Inventory	Create PI document, count, post difference	Stock adjusted in MMBE | FI document via OBYC INV
ML Costing Run	Execute CKMLCP for period and review CKM3N	Actual cost differs from standard — revaluation entry posted

14. Project Summary

- Configured complete three-plant enterprise structure for DMAG across Düsseldorf, Dortmund and Stuttgart
- Built material master for ROH, HALB and FERT types with custom number ranges and valuation classes
- Designed custom pricing procedure ZDM01 using condition technique with gross price, discount and freight
- Implemented four-level PO release strategy using classification-based approval workflow
- Activated Material Ledger with EUR and USD parallel currencies across all plants
- Configured complete OBYC account determination covering BSX, WRX, PRD, GBB, FRE, KON and INV
- Executed MRP Live with reorder point planning and PIR-based demand signals
- Configured and tested STO, Consignment and Subcontracting with BOM
- Implemented Batch Management with Split Valuation for Steel Sheet by quality grade
- Executed complete Physical Inventory cycle with FI difference posting